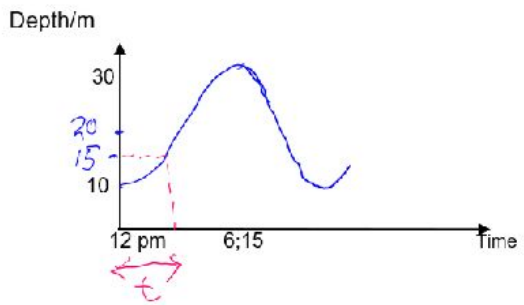
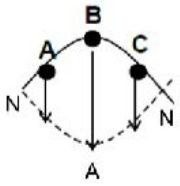
8(a)(i)	SHM is an oscillatory motion (of an object) where the acceleration is (directly) proportional to displacement from the equilibrium position/a certain fixed pt & acceleration is always directed towards the equilibrium position/the fixed pt.	[1] [1]
8(a)(ii)	Deduce ½ period = 6.15 pm – 12 pm period T = 12.5 hours Deduce amplitude x_o = ½ (30 – 10) m x_0 = 10 m Deduce equation of motion is: $x = -x_o \cos \omega t$ where t = 0 corresponds to 12 pm $x = -10 \cos\frac{2}{12.5}t$ Marking: Allow ecf for values of x_0 & T  Since minimum depth required = 15 m, the earliest entrance is at depth of 5 m below equilibrium position where equilibrium position is at depth of 20 m, earliest entrance occurs at x = - 5 m Thus, $-5 = -10 \cos\frac{2}{12.5}t$ where t = time in hours after 12 noon $0.5 = \cos^{-1}\frac{2}{12.5}t$ t = 2.08 hrs = 2 hrs + 0.08(60) mins = 2 hrs + 4.8 mins = 2 hrs + 5 mins earliest time = 12 noon + 2 hrs and 5 mins = 2.05 pm	[1,ecf] [1] [1] [1, ecf]
8(a)(iii)	The tidal motion is a simple harmonic motion.	[1]
8(b)(i)1.	Period T = 20 ms {read-off the graph} f = 1/T = 50 Hz	[1] [1]

8(b)(i)2.	Sketch of displ-position graph for fundamental mode of vibration}  Within each loop: amplitude is different for every point (except at nodes) Since max acceleration∝ amplitude, {do not accept: *displacement*} Max acceleration at all points is not the same. Marking: Award last mark only if first 2 marks are scored.	[1] [1] [1]
8(b)(ii)1.	resonance	[1]
8(b)(ii)2.	50 Hz (ecf, bi) corresponding to max amplitude { or equivalent }	[1,ecf]
8(b)(ii)3.	Damping is the effect due to a gradual loss of energy from an oscillating system due to resistive forces like friction or air resistance, which causes the amplitude of the oscillations to decrease over time 2nd graph: 1. approximately same (or slightly lower) resonant frequency & 2. sharpness of resonance reduced ie "obviously broader peak". 3. smaller amplitude at every freq, (excluding pt 4); hence curves must not cross passes through same point on vertical-axis as original curve 4. passes through same point on vertical-axis as original curve {ie at f = 0}	[1] [1] [1] [1]